

Problem 10.11

Ernie makes deposits of 100 at time 0, and X at time 3. The fund grows at a force of interest

$$\delta_t = \frac{t^2}{100}, \quad t > 0.$$

The amount of interest earned from time 3 to time 6 is X . Calculate X .

Solution

The accumulation function from time 0 to time t is

$$a(t) = \exp \left(\int_0^t \delta_s ds \right).$$

Substituting the force of interest gives

$$a(t) = \exp \left(\int_0^t \frac{s^2}{100} ds \right) = \exp \left(\frac{t^3}{300} \right).$$

At time 3, the initial deposit has accumulated to

$$100a(3) = 100e^{27/300} = 100e^{0.09} = 109.4174284.$$

After the deposit of X , the fund value at time 3 is

$$A(3) = 100e^{0.09} + X.$$

The accumulation factor from time 3 to time 6 is

$$\exp \left(\int_3^6 \frac{t^2}{100} dt \right) = \exp \left(\frac{6^3 - 3^3}{300} \right) = e^{0.63}.$$

Therefore,

$$A(6) = (100e^{0.09} + X) e^{0.63}.$$

The interest earned from time 3 to time 6 is $A(6) - A(3)$, and this amount is given to be X . Hence,

$$(100e^{0.09} + X) e^{0.63} - (100e^{0.09} + X) = X.$$

Factor the left-hand side:

$$(100e^{0.09} + X) (e^{0.63} - 1) = X.$$

Solving for X gives

$$X = \frac{100e^{0.09}(e^{0.63} - 1)}{2 - e^{0.63}}.$$

Therefore,

$$\boxed{X = 784.593081}.$$